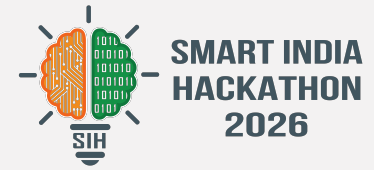


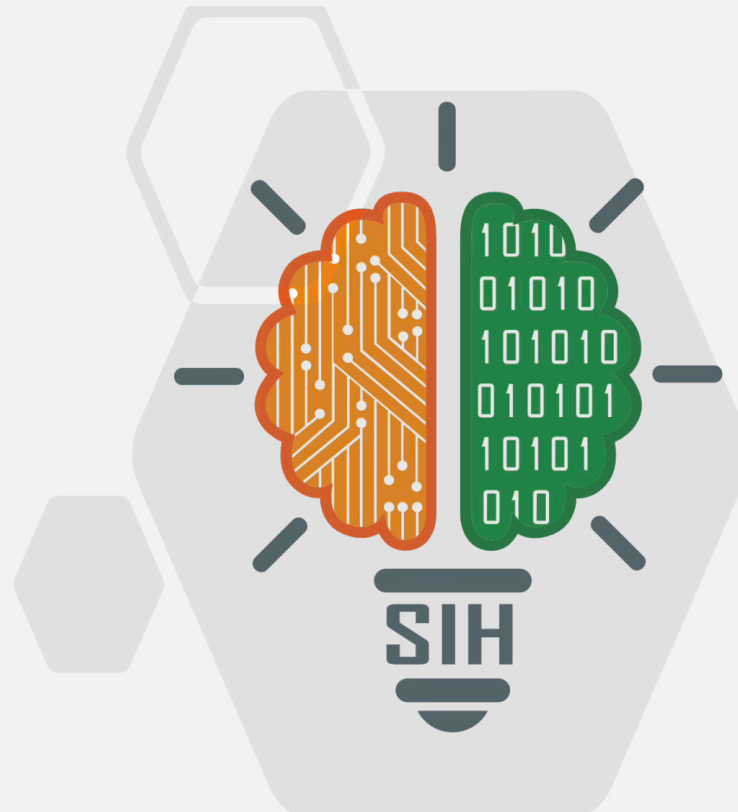
SMART INDIA HACKATHON 2026



ATITHYA

Hospitality every village already has. A market it has never reached.

- **Problem Statement ID** – 26204
- **Problem Statement Title** – Student Innovation
— A solution/idea that can boost the current situation of the tourism industries including hotels, travel and others
- **Theme** – Travel & Tourism
- **PS Category** – Software
- **Team ID** – To be assigned
- **Team Name (Registered on portal)** – **Claude Can Code**



ATITHYA — Proposed Solution

ROOT CAUSE Village homestays never get listed because an OTA must send staff to visit and write up each one — **about ₹2,500 per listing.**

DETAILED EXPLANATION OF THE PROPOSED SOLUTION

A village host sends one WhatsApp message. Our agent interviews them, writes the listing, publishes it — then sells it to tourists.

WHATSAPP · AGENT ↔ HOST	
ATITHYA Namaste! Aap kya offer karte hain?	₹30 to onboard a listing, against ₹2,500 for an OTA field visit
HOST Do kamre hain, khana bhi...	4 agents — onboarding, verification, planning, re-planning — under one LangGraph supervisor
ATITHYA Photo bhejiye — listing ban gayi.	0 apps to install, forms to fill or English required of the host

HOW IT ADDRESSES THE PROBLEM

A person must visit and write up each listing

→ **The agent interviews the host on WhatsApp instead**

Listing means typing English into an app

→ **The host speaks Hindi; the agent writes the listing**

Tourists juggle four apps and no safety signal

→ **One thread plans, books, warns and re-plans**

States cannot see where rural tourism spend lands

→ **Bookings roll up into a district-level dashboard**

INNOVATION AND UNIQUENESS OF THE SOLUTION

CAPABILITY	MAKEMYTRIP / YATRA	OYO / AIRBNB	AI TRIP PLANNERS	ATITHYA
Lists unorganised supply	No	Limited	No	Core focus
Onboards non-tech-literate operators	Manual, sales-heavy	App-based, English-first	Not applicable	WhatsApp, any language
Persistent memory across sessions	No	No	Resets each chat	Accumulates full history
Real-time safety & scam advisories	No	No	No	Yes
Policy-level analytics for states	No	No	No	Unique to Atithya

TECHNICAL APPROACH

TECHNOLOGIES TO BE USED

ORCHESTRATION **LangGraph** supervised agent graph

BACKEND **Node.js + Express** booking APIs

DATA **PostgreSQL + pgvector** checkpoints

CHANNEL **WhatsApp Cloud API**, no install

INTERFACE **React** live dashboards

GROUNDING **RAG** over verified listings

METHODOLOGY AND PROCESS FOR IMPLEMENTATION

SUPERVISOR LangGraph state machine — routes every step, retries on failure, checkpoints state

01 ONBOARDING

Host conversation → structured listing

02 VERIFICATION

Anomaly and safety screen before publish

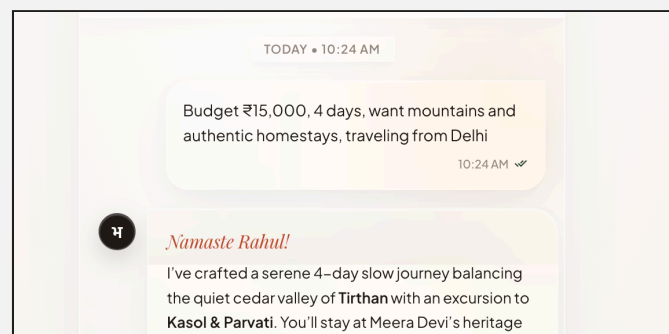
03 PLANNING

Budget and interests → itinerary → booking

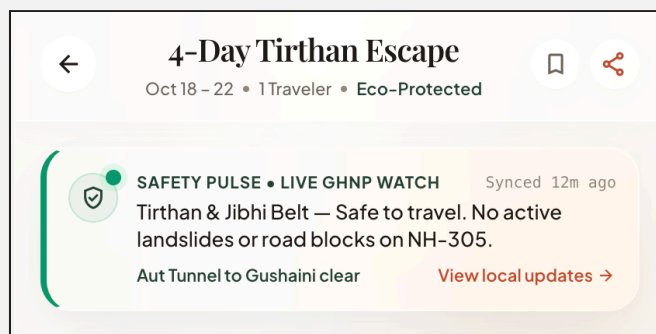
04 RE-PLANNING

Rebuilds the trip on weather or closures

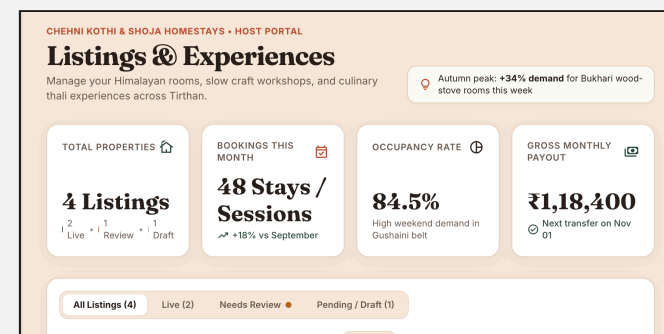
WORKING PROTOTYPE Built and running today — tourist planning agent, live trip view and host portal.



01 Planning agent — budget in, itinerary out



02 Live trip — real-time safety pulse



03 Host portal — occupancy and payouts

Roadmap — Weeks 1–4 onboarding agent, 50 pilot hosts · Weeks 5–8 booking and payments · Weeks 9–12 verification, re-planning, government dashboard

FEASIBILITY AND VIABILITY

ANALYSIS OF THE FEASIBILITY OF THE IDEA

Every component ships today — WhatsApp Cloud API, LangGraph, pgvector. No new research required.

The channel already exists: WhatsApp is on the phone of every operator we onboard. No app adoption problem.

Supply acquisition is software, not headcount. Fifty hosts or fifty thousand run on the same stack; a new district is configuration, not hiring.

REVENUE MODEL

Commission	10–15% per confirmed booking, host and tourist side	P1
Host Pro tier	₹299/month — pricing copilot, analytics, priority slots	P2
Partner placement	Transport, guides and artisan cooperatives	P2
B2G licence	State board subscription for scheme-outcome tracking	P3
Demand insights	Anonymised regional demand data to chains and DMCs	P3

UNIT ECONOMICS — COST TO PUT ONE LISTING ONLINE

OTA field team visit, photograph, write listing		₹2,500
Atithya agent one WhatsApp conversation		₹30
83x cheaper per listing. Onboarding stops being a headcount cost, so rural supply finally clears the bar to come online.		

WHAT THAT UNLOCKS

₹432 Platform revenue per booking, at 12% commission	₹6,336 Host's new monthly income at two bookings	₹2.1 cr Year-1 revenue on ₹17.3 cr GMV, 2,000 listings
Indicative projections, modelled from public data.		

POTENTIAL CHALLENGES AND RISKS

Listing quality and fraud in unverified supply

Host drop-off after the first conversation

Thin demand in a new district

Dependence on a single messaging platform

STRATEGIES FOR OVERCOMING THESE CHALLENGES

Verification agent screens anomalies before a listing goes live

Agent re-engages over WhatsApp with booking alerts and pricing nudges

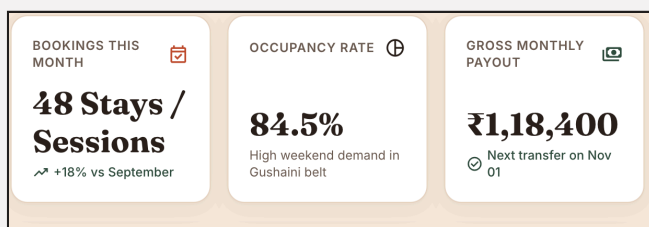
Launch district by district around existing high-traffic anchors

Channel-agnostic agent layer; IVR and SMS fallback for low-connectivity regions

IMPACT AND BENEFITS

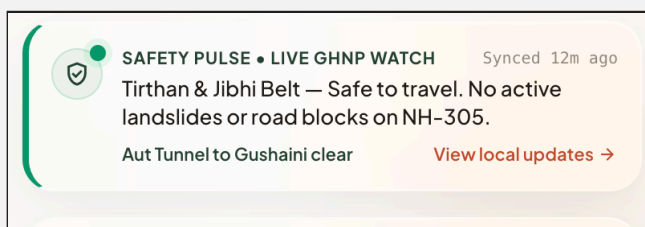
POTENTIAL IMPACT ON THE TARGET AUDIENCE

HOST · OPERATOR



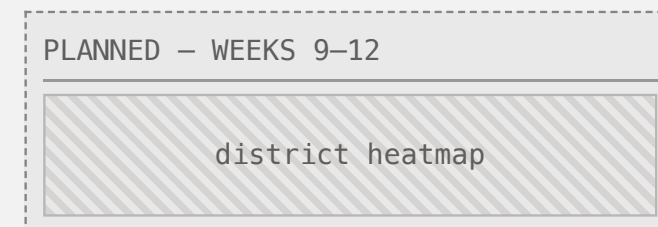
A homestay owner with no website and no English gets a live, bookable listing, a pricing copilot and a monthly payout.

TOURIST



One surface for planning, booking, budget, safety pulse and mid-trip re-planning — in place of four disconnected apps.

STATE TOURISM BOARD



Sees which districts are absorbing rural tourism spend and which are not — scheme outcomes, not scheme outlays.

BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL)

SOCIAL

Income reaches women- and youth-run households in villages that no OTA lists. Language stops deciding who is allowed to sell to a tourist.

ECONOMIC

SBI Research puts national homestay need at roughly 3 million units, with ₹1,500 cr of MUDRA Shishu lending already opened for them. Atithya turns funded rooms into booked rooms.

ENVIRONMENTAL

Routing demand to underused destinations eases crowding at the same few sites and spreads footfall across a wider set of districts.

RESEARCH AND REFERENCES

9.02 mn

Foreign tourist arrivals, 2025 — down 9.4% year on year, but up 4.25% excluding Bangladesh^{1, 2}

₹2,438 cr

Ministry of Tourism allocation, Union Budget 2026–27³

~3 mn

Homestays estimated to be needed nationally; ₹1,500 cr MUDRA Shishu opportunity⁵

₹2,208 cr

Sanctioned across 53 Swadesh Darshan 2.0 projects⁶

DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

- 1 Ministry of Tourism — India Tourism Data Compendium 2025 (FTA tables, ch. 2). [India Tourism Data Compendium 2025 \(PDF\)](#); live series at data.tourism.gov.in
- 2 Business Standard — FTAs fell 12.4% in Apr–Sep 2025 to 6.18 mn; 2019 peak was 10.93 mn. business-standard.com
- 3 Union Budget 2026–27 — Notes on Demands for Grants, Demand No. 99, Ministry of Tourism. indiabudget.gov.in/doc/eb/sbe99.pdf
- 4 Union Budget 2025–26 — MUDRA loans for homestays, top 50 destinations in challenge mode, streamlined e-visa. business-standard.com
- 5 SBI Research, reported by ANI — ~3 mn homestays needed; ₹1,500 cr Shishu-category opportunity, up to ₹8,250 cr at Kishore. tribuneindia.com
- 6 PIB — Swadesh Darshan 2.0: 53 projects, ₹2,208.27 cr sanctioned; 36 CBDD projects, ₹648.11 cr. pib.gov.in/PRID=2212575
- 7 PIB — Promotion of rural tourism; rural circuit under Swadesh Darshan; Incredible India B&B classification. pib.gov.in/PRID=2220116
- 8 Swadesh Darshan 2.0 — scheme guidelines, Ministry of Tourism. sd2.tourism.gov.in/AboutUs.aspx
- 9 LangGraph — multi-agent supervisor pattern and persistent checkpointing. langgraph/concepts/multi_agent
- 10 WhatsApp Business Cloud API reference: developers.facebook.com/docs/whatsapp/cloud-api. pgvector: github.com/pgvector/pgvector